THE CLEANER

Game Design Document

Platform: Game Boy Advance (GBA) via Delta Emulator

Target: iPad Pro (3rd Generation)

Status: Living Document — Subject to Change

Document Note

All character names, locations, story beats, and mechanics described in this document are working drafts. Names, backstories, and specific details are placeholders and will be revised during development. The mechanical systems and structural design are the stable foundation; narrative details are subject to change.

1. Project Overview

Title: The Cleaner (working title)

Genre: Top-down semi-open world crime / stealth

Platform: Game Boy Advance — playable via Delta emulator on iPad Pro

Language: C with devkitARM toolchain and libgba / libtonc

Developer: Solo

Setting: New York City, 1973

Concept

The Cleaner is a top-down, semi-open world GBA game in the tradition of the original Grand Theft Auto series, with a tone drawn from classic mafia cinema — The Godfather, The Irishman, Scarface, and The Sopranos. The player takes the role of Vincent Marra, an 18-year-old released from Rikers Island who is recruited by the Cavallo crime family's consigliere. What begins as a series of jobs to prove himself gradually unravels into a conspiracy that reframes everything Vincent has done.

The game emphasizes stealth, social navigation, and moral consequence over combat. Replayability comes from a mission pool system, a per-contact reputation ladder, and four distinct ending paths determined entirely by the player's choices throughout.

Core Design Pillars

- Small world, deep systems — five zones replace a large open world; every location has meaning
- Reputation over combat — relationships and trust are the primary currency
- Moral ambiguity — no ending is labeled good or bad; the player arrives at one naturally
- Replayability — mission pools, branching contacts, and rep-gated content reward multiple playthroughs
- Hardware respect — every design decision works within GBA constraints

2. Technical Constraints (GBA Hardware)

Screen

Spec	Value
Resolution	240 x 160 pixels
Refresh rate	~60 fps
3D support	None — Mode 7 scaling/rotation tricks only
Tile size	8 x 8 pixels

Memory & CPU

Spec	Value
CPU	ARM7TDMI at 16.78 MHz
Work RAM	256 KB external + 32 KB internal
ROM size	Up to 32 MB (target: 4–8 MB)
Max sprites	128 on screen (max 32 per scanline)
Sprite sizes	8x8 up to 64x64 pixels

Color

Spec	Value
On-screen palette	256 colors simultaneously
Total available	32,768 colors
Sprite palettes	16 palettes x 16 colors each

Audio

- 2 x 8-bit PCM channels — no built-in MIDI
- Software mixing required — use Maxmod library
- Audio processing eats CPU budget — keep music loops efficient

Zone Architecture

Zones are separate screens connected by transition points. Each zone loads independently, keeping RAM usage per zone within budget. At 8x8 pixel tiles, each zone is approximately

60x40 tiles per screen — roughly 180–240 tiles wide and 80–120 tiles tall at 3–4 screens wide and 2–3 screens tall per zone.

3. Control Scheme

Button	Context	Action
D-pad	Always	Move Vincent in 8 directions (speed varies by mode)
A button	Near NPC / building	Interact — talk, enter, pick up, confirm dialog
A button	No interactable nearby	Use equipped item — fire weapon, deploy tool
B button	Normal zone	Sprint — move faster, raises suspicion in guard sightlines
B button	Stealth zone	Sneak — slower, narrows noise radius, reduces detection
L bumper (hold)	Always	Open item wheel (4 slots) — D-pad selects, release to use
R bumper (hold)	Always	Map overlay — neighborhood, objectives, contact locations
Start	Always	Pause menu — resume, settings, quit to title
Select	Always	Open journal / casebook (tabbed)

Context-Sensitive B Button

B switches automatically based on tile flags — the same system used for collision detection. Stealth zones are tiles with a stealth flag set. A small icon beside Vincent indicates current mode (run icon = sprint available, eye icon = sneak active). The transition is instant with no animation interrupt.

Item Wheel (L Bumper)

- 4 item slots maximum — up/down/left/right on D-pad while L is held
- Loadout selected at safehouse or social club before heading out — cannot swap mid-mission
- Empty slots are greyed out — no penalty for carrying fewer items
- Forces planning: camera vs. spare key vs. distraction item vs. bribe envelope

Journal / Casebook (Select)

Tab	Contents
Case log	Running narrative recap in Vincent's voice — terse, first person. Auto-updates after missions.
Active mission	Current objectives, location, known complications. Updates mid-mission.

Contacts	Rep meter (0–3) per contact, missions completed, brief character note expanding with rep.
NPC profiles	Unlocked entries for named NPCs — name, role, last known location, gathered notes.
Finances	Current cash, mission payouts, tool purchase history.

4. World & Map Layout

The neighborhood is divided into five zones connected by transition points. No zone is directly accessible from all others — routing through intermediate zones is intentional design. The camera centers on Vincent at all times unless he reaches the zone edge.

Zone Connections

Connection	Transition point	Notes
Zone A ↔ Zone B	Main Street crossing	Wide, always accessible — primary route between hub and commercial
Zone A ↔ Zone D	Back alley	Narrow — Caruso's patrol crosses here. Learning his timing is essential.
Zone B ↔ Zone C	Dock road (gated)	Unlocks after story beat 1. Guard on duty at night after beat 2.
Zone D ↔ Zone C	Canal bridge	Partially blocked early. Fully opens after beat 1.
Zone D ↔ Zone E	Private road	Requires invitation (Sal rep 2+) or disguise. Unlocks beat 1.
Zone A ↔ Zone C	No direct route	Must route through B or D — keeps active zones relevant throughout.

Zone A — Social Club + Residential (Hub)

The starting zone. Wide enough to breathe — a main street running east-west, two side streets, and alleyways behind brownstones. Feels lived-in and safe early on. Atmosphere shifts after story beat 1.

Building	Access	Notes
The social club	Always open	Hub. Sal is here. Atmosphere shifts per act.
Apartment block A	Always open	Ambient. Three enterable units with rotating NPC dialog.
The hardware store	Always open	Functional. Buy tools before heading out.
The barbershop	Always open	Ambient. Good eavesdropping. One contact meets you here once.
The hidden basement	Unmarked alley door	No mission sends you here. Pure lore reward for explorers.

Zone B — Commercial Strip

The busiest zone — most mission locations draw from here. A long commercial block with a side street cutting through the middle. Dense civilian foot traffic creates natural cover. Busiest during day, quieter at night.

Building	Access	Notes
Rizzo's Diner	Always open	Lowers suspicion passively. Back office is a mission location.
The pawn shop	Always open	Sell items for cash. Fence for certain mission objects.
The laundromat	Sal rep 2+	Sal's front. Back room used in several missions.
The pharmacy	Always open	Ambient. NPC hints at Zone C.
The parking structure	Always open	Elevated sightline. Used in one surveillance mission.

Zone D — Back Streets + Tenements

Tighter, more claustrophobic than Zone A. Narrower streets, more alleyways. A canal runs along the southern edge leading to the bridge into Zone C. Caruso does regular passes through this zone — his patrol route is learnable.

Building	Access	Notes
Dena's apartment	After mission 1	Key story conversations at rep 2 with Dena.
Tenement block B	Always open	NPC tracks Caruso's movements for you at rep 1.
Enzo's payphone	Beat 1 (outdoor)	Interaction point only — no building. Unlocks beat 1.
The canal walkway	Always open	Open space used in two missions as extraction route.
The lockup garage	Enzo rep 2+	Stores mission-critical items in act 3.

Zone C — Waterfront (Unlocks Beat 1)

Wide open loading areas, chain-link fencing, long sightlines. Exposed and industrial — the opposite feel to Zone D. Fewest hiding spots, highest guard density. Final mission plays out entirely here.

Building	Access	Notes
Warehouse A	Partial beat 1, full beat 2	Corkboard showing NPC connections — lore reward.
Warehouse B	Final story beat	Mission location in act 3.
Loading dock office	Always (after zone opens)	One NPC — bribeable for patrol timing info.

The docked boat	Always (after zone opens)	Used in one Enzo mission and the final act.
The gatehouse	Always (after zone opens)	Transition back to Zone B. Guard stationed here at night.

Zone E — Affluent District (Unlocks Beat 1)

Wider streets, manicured stoops, fewer civilians but more private security. Stark contrast with other zones — Vincent doesn't belong here and the environment communicates that. Connects to Zone D via a private road.

Building	Access	Notes
Campaign headquarters	Always (after zone opens)	Mayor's public face — staff, volunteers. Front for the back room.
The mayor's residence	Mayor rep 2+	Quieter, personal. Real conversations happen here.
The private club	Disguise or forged invite	Several powerful NPCs meet here. One surveillance mission.
The law office	Always (after zone opens)	Mayor's counsel. Hints at money trail if Dena rep is high.
The park	Always (after zone opens)	Open space. One outdoor surveillance mission. Exposed.

5. Characters

Note on names and backstories

All character names and backstories are working placeholders. They will be revised during development. The roles, relationships, and mechanical functions described below are stable design.

Vincent Marra — The Player Character

- Age: 18–19 at game start
- Background: Released from Rikers Island after 1–2 years following a conviction at age 17 (transferred from juvenile detention per NYC law)
- Starting rank: Associate — non-member working with the family, proving himself
- Character arc: Moves from associate toward soldier across the story — though which version of that journey completes depends entirely on the player
- Key trait: He handled himself inside. Kept quiet. Didn't fold. That's why the lawyer was waiting.

The Lawyer — Consigliere

- Role: Consigliere to the Don — advisor, legal expert, trusted confidant
- Function: Narrative figure at opening and story beats; also a contact with the most cerebral mission type
- Mission type: Legal interference — document interception, witness coaching, institutional sabotage
- Key trait: Knows the full picture from the start. Every scene with him is retrospectively interesting.
- Crucial reveal: He arranged Vincent's recruitment deliberately — Vincent's time in Rikers was engineered
- Access: Summons Vincent rather than being freely visited. Appears at every story beat.

Sal Cavallo — Underboss

- Role: Underboss — second in command, oversees daily operations
- Function: Primary contact for field work — extractions, relocations, escorts, physical jobs
- Tone: Warm, paternal on the surface. Increasingly complex as the story develops.
- Rep rewards: Rep 2 unlocks the laundromat back room. Rep 3 unlocks the waterfront and family residence.
- Key tension: The gap between who Sal appears to be and who he actually is drives much of the drama.

Dena Ricci — Bookkeeper

- Role: Family associate — keeps the books, holds no rank, wants none
- Function: Contact for information work — surveillance, evidence, intercepts
- Unlocked: After opening mission (Vincent relocates her)
- Rep rewards: Rep 2 unlocks key story conversation at her apartment. Rep 3 gives Vincent the full truth.
- Key role: Holds the evidence that reframes the entire story at beat 3.

Detective Caruso — The Tail

- Role: NYPD plainclothes — entirely outside the family structure
- Function: Contact for clean work — disruption, evidence gathering, institutional exposure
- Tone: Not corrupt, not stupid. Has been watching the neighborhood for months.
- Rep rewards: Rep 3 opens the justice ending path. High Caruso rep burns standing with Sal.
- Patrol: Crosses the Zone A / Zone D back alley regularly — his route is learnable.

Enzo — The Fixer

- Role: Freelance — operates outside family structure, no allegiance
- Function: Contact for high-risk, high-reward work. Unlocks at story beat 1.
- Tone: Cold, mercenary. Takes jobs the family pretends don't exist.
- Rep consequences: Taking his work passively drains rep with Caruso. High Enzo rep unlocks assassination ending.
- Access point: A payphone in Zone D. No building.

The Mayoral Candidate

- Role: Reform mayoral candidate — charismatic, partially compromised
- Relationship to family: Taking money, asking few questions. Genuinely believes parts of his platform.
- Function: Contact for political operations missions. Instructions always come through his assistant.
- Location: Campaign HQ in Zone E
- Beat 2 outcome: Elected, arrested, or assassinated — depends on missions completed and rep profile
- Three paths persist after outcome: Continue supporting him / bring him to justice / have him removed

The Don — Boss of the Cavallo Family

- Role: Head of the family. All major decisions flow from him.
- Presence: Never seen in acts 1–2. Referenced in dialog, behavior of others, the way rooms change.
- First appearance: After beat 3 — the lawyer arranges a single mandatory meeting. No dialog choice.
- Content: What the Don says depends on Vincent's rep profile across all contacts.
- Ending A only: Vincent returns after the final mission. The making ceremony. The Don nods. That is enough.

Capos — Zone-Level Family Presence

Each zone has a named capo who reacts to Vincent's reputation rather than giving missions. They govern territory. High Sal rep means capos are cooperative and occasionally informative. Enzo-heavy play makes some capos hostile.

Zone	Tone	Notes
Zone A	Wary → warming	Oldest capo. Wary of Vincent initially. Useful source of neighborhood history.
Zone B	Transactional	Businesslike. Cares about results not relationships. Most neutral.
Zone D	Ambitious → hostile	Youngest capo. Dislikes Vincent's independence. Becomes hostile if Enzo rep is high.
Zone C	Taciturn	Controls docks. Rarely seen before act 2. Cooperation required for endings A and C.

6. Cavallo Family Hierarchy

Rank	Character	Role in game
Don / Boss	The Don (name TBD)	Mythic presence. One scene post-beat 3. Full appearance in ending A only.
Underboss	Sal Cavallo	Primary contact for field work. Second most important character.
Consigliere	The Lawyer (name TBD)	Recruiter, narrative anchor, cerebral mission contact.
Capo x4	One per zone (names TBD)	Reactive NPCs — not contacts. Reflect Vincent's reputation in their zone.
Soldiers	Named NPCs throughout	Mission targets, obstacles, or reluctant allies. Vincent is not yet one of them.
Associate	Vincent Marra (opening)	Where Vincent starts. Trusted enough for work, not trusted enough to know everything.

Vincent's Rank Progression

Story point	Status
Opening	Associate — one job to prove himself. No guarantees.
After beat 1	Trusted associate — recognized by capos in zones A and B.
After beat 2	Candidate for soldier — the question is officially on the table.
After beat 3	Ambiguous — the setup revelation complicates everything.
Ending A	Made man — the ceremony. Embedded, permanent, complicit.
Endings B/C/D	Never made — each for different reasons.

7. Story Structure

Story beat note

The current structure has 4 story beats plus an opening. More beats will likely be added during development. Mission count thresholds are working estimates and will be tuned during playtesting.

Opening — Release Day (Fixed)

Vincent walks out of Rikers. The lawyer is waiting. A brief conversation — no pressure, just an observation about Vincent's options. One introductory mission: relocate Dena Ricci to a safehouse. Teaches movement, interaction, and the suspicion system. Dena becomes a contact. Zones A and B accessible. Estimated time to freeplay: 10–15 minutes.

Story Beat 1 — The Introduction (After 4 side missions)

The lawyer reappears and introduces Vincent to Sal at the social club. Zone E unlocks — the mayoral candidate becomes a contact. Enzo surfaces in Zone D. The dock road to Zone C partially opens. Vincent stops being a new face. Capos begin to acknowledge him. Enzo's payphone becomes active.

Story Beat 2 — Election Night (After 10 side missions)

The mayoral election lands. Outcome depends on missions completed for the candidate and overall rep profile. Three possible outcomes, each reshaping the final act:

Elected

Vincent did enough campaign work. The mayor wins. Zone E becomes more guarded. The family is stronger than ever. But the mayor — now with real power — begins distancing himself from Sal. A new internal threat emerges.

Arrested

Caruso or Dena's missions gave someone enough to act on. The mayor is taken on election night. The family scrambles. Sal goes quiet. The lawyer makes calls. Vincent becomes a loose end from both sides.

Assassinated

Sufficient Enzo rep and a specific act 1 mission triggered this. The mayor never reaches the ballot count. Zone E locks permanently. No one discusses who gave the order. Vincent knows.

Story Beat 3 — The Revelation (After 16 side missions)

Dena surfaces with the complete picture — the money trail, the family's fingerprints on the campaign, and the name of whoever inside the family arranged Vincent's conviction at 17. It was not random. He was set up, watched, and recruited deliberately. The lawyer knew. Whether Sal knew is the question the ending answers differently per rep profile. Zone C fully unlocks. The final act approaches.

Story Beat 4 — The Thread (After 20 missions or manually after beat 3)

The final mission. Its shape depends entirely on the rep profile built across the whole game and which mayor outcome occurred at beat 2. Four distinct final missions, no two playthroughs guaranteed to reach the same one. The neighborhood becomes the stage for a quiet, precise, and irreversible ending.

8. Ending Paths

No ending is labeled good or bad. The player arrives at one naturally based on who they spent the game working for. All four endings are legitimate resolutions of the story.

Ending	Name	Required rep	Outcome
A	The Loyalist	High Sal	Vincent protects the family. He stays. The neighborhood survives. He becomes a made man. Nothing changes — and that is the point.
B	The Conscience	High Caruso + Dena	Vincent delivers everything to Caruso. The family takes a hit. He burns every bridge and leaves for good. The cleanest ending morally, the costliest personally.
C	The Professional	High Enzo	Vincent removes the problem. No trial, no exposure. He gets paid and feels nothing. Earned only through consistent choice. Darkest ending.
D	The Ghost	Balanced across all	Vincent knows too much to stay and too little to act. He disappears. The neighborhood continues without him. Most ambiguous, hardest to reach.

9. Core Gameplay Systems

Reputation System

Each of the five contacts (lawyer, Sal, Dena, Caruso, Enzo, mayor) has a separate reputation score on a 0–3 scale. Rep is gained by completing missions cleanly — no suspicion triggered, objectives fully met. Partial completions give partial rep. The player cannot max all contacts in one playthrough.

Level	Symbol	Effect
0	○○○	Cold — one mission available, no building access, minimal dialog.
1	●○○	Standard — full mission pool unlocked, contact opens up.
2	●●○	Trusted — new building access, harder missions, personal story revealed.
3	●●●	Full trust — contact's arc completes, unique ending path available.

Mission Pool System

Each contact has a pool of 8–12 authored missions stored in ROM. Each playthrough draws from the pool randomly. Each mission has three variable slots: a target location, a target NPC, and a complication modifier. Same mission type, different feel each run.

- Location variables: drawn from 4–5 buildings per zone
- Complication variables: guard presence doubled, time limit active, Caruso is watching, rain reduces NPC visibility, target is already spooked
- A single playthrough sees ~12 missions from a pool of 30+ — leaving substantial content for replays

Stealth System

- Enemy detection cones — visible, blocked by dense tile cover
- Three alert states: unaware, suspicious, hunting — drops over time if player breaks line of sight
- Noise radius — affected by movement speed (sprint increases it, sneak decreases it)
- Suspicion meter — fills near guards or when civilians witness incidents, drains when player leaves area or uses cover actions (sitting at diner, etc.)
- Stealth zones — flagged tiles that automatically switch B button from sprint to sneak

Soft Story Triggers

Story beats trigger based on total missions completed — not which specific missions. The player chooses order freely. The story catches up to them rather than leading them. This creates genuine non-linearity without requiring complex branching mission logic.

Building Access

Type	Examples	Notes
Always open	Diner, hardware store, barbershop	Ambient or functional. No gate. Texture and utility.
Rep-gated	Laundromat, Dena's apartment	Opens when contact rep threshold is reached.
Story-gated	Warehouse A/B, waterfront	Opens at specific story beats regardless of rep.
Hidden	The basement room	No mission sends you here. Rewards pure exploration.

Tool Inventory

Four tools maximum per mission, selected at the safehouse or social club before heading out. Cannot swap mid-mission. Forces strategic planning.

Tool	Function
Camera	Photograph targets for evidence missions.
Spare key set	Bypass a locked door once per mission.
Bribe envelope	Pay off a civilian witness.
Distraction item	Noise draw — pulls a guard off their route.
Disguise	Changes NPC recognition for a limited time.
Weapon (act 3+)	Lethal option. Available only in certain missions.

Consequence Flags

Binary flags track key choices across missions — 3–4 boolean variables that the final mission reads. Did Caruso see Vincent's face in mission 2? Is Dena cooperative or hostile? Was the ledger fully destroyed? Each flag opens or closes one path in the ending. Simple to implement in C, feels meaningful to the player. No explicit morality meter is shown.

10. Contact Mission Types

Contact	Mission type	Sample missions
Sal Cavallo	Field work	Move witness, intercept courier, extract burned soldier, escort Sal's nephew
Dena Ricci	Information	Photograph meeting, retrieve ledger page, plant listening device, tail a mark
The Lawyer	Legal interference	Document interception, witness coaching, evidence suppression, institutional sabotage
Detective Caruso	Clean work	Gather evidence, sabotage a delivery, surveil a city official, plant anonymous tip
Enzo	High risk	Destroy evidence before sweep, recover stolen property, disrupt a meeting, waterfront job
The Mayor	Political ops	Deliver envelope to union rep, intercept a journalist's source, disrupt rival event, recover photos

11. Development Notes

Toolchain

- Language: C (primary), C++ (if needed)
- SDK: devkitARM + libgba or libtonc
- Audio: Maxmod library
- Target output: .gba ROM file
- Emulator: Delta (iPad Pro 3rd gen via TestFlight or AltStore)

Developer Background

Developer is proficient in C, C++, Java, JavaScript, Python, R, and HTML/CSS. The primary learning curve is GBA-specific hardware knowledge — memory-mapped I/O registers, tile systems, sprite OAM — not the C language itself.

Recommended Build Order

- 1. Development environment setup — devkitARM, emulator testing pipeline
- 2. Hello World — sprite on screen, button input reading
- 3. Movement system — 8-directional movement, camera centered on player
- 4. Tile map — Zone A layout, collision detection via tile flags
- 5. NPC system — simple patrol AI, interaction trigger
- 6. Stealth system — detection cones, alert states, suspicion meter
- 7. Mission system — mission struct, one complete mission end-to-end
- 8. Reputation system — per-contact rep tracking, rep-gated content
- 9. Item wheel — L bumper wheel, loadout selection screen
- 10. Journal / casebook — Select button UI, tabbed navigation
- 11. Zone transitions — loading/unloading zones, transition point triggers
- 12. Story beat system — mission count tracking, beat triggers
- 13. Audio — Maxmod integration, ambient music per zone
- 14. Content pass — remaining zones, buildings, mission pool population
- 15. Ending system — consequence flags, final mission branching

Key Constraints to Keep in Mind

- 256 KB work RAM — zone data, mission structs, and NPC data all compete for the same pool
- 128 sprite limit — 32 per scanline is the real constraint in dense scenes

- Tile-based backgrounds — art must be designed as 8x8 tile sheets
- No runtime AI generation — all mission content must be authored and compiled into ROM
- Audio uses CPU — keep music loops efficient, test performance impact early

Replayability summary

A single playthrough completes roughly 12 side missions out of a pool of 30+, maxes 1–2 contacts out of 5, explores most but not all buildings, and reaches one of four endings. A second playthrough with different contact priorities produces a meaningfully different experience — different missions, different building access, different ending, different picture of what actually happened.